Extended Abstract

Motivation Loco-manipulation—the combined ability to move through an environment while interacting with objects—is a critical skill for humanoid robots assisting humans in household and manufacturing tasks. Recent advances have shown promising results in learning whole-body controllers that imitate human motion, enabling robots to perform diverse behaviors such as walking, dancing, and kicking. However, these efforts largely neglect object interaction, which is essential for real-world applications. Meanwhile, teleoperation systems have been developed to teach robots object manipulation. These systems use headsets to provide egocentric visual feedback as human operators guide robot arms, often in tabletop settings. While effective for collecting paired egocentric observations and robot actions, these systems are limited: they typically do not capture lower-body movements, and thus cannot support tasks requiring full-body coordination—such as walking while carrying objects or crouching to pick items from the floor. Many real-world tasks—like moving boxes, lifting chairs, or rearranging furniture—require this kind of full-body loco-manipulation. Teaching these skills presents two key challenges: How can we efficiently collect training data for humanoid loco-manipulation? Extending teleoperation to capture full-body motion is tedious, hardware-intensive, and difficult to scale. In contrast, there is a wealth of human-object interaction data—including motion capture, third-person videos, and egocentric recordings—as well as generative models from character animation (e.g., motion diffusion models from language) that can synthesize diverse and scalable training examples. If properly adapted, these can be used to train humanoid loco-manipulation policies. To address this, we first retarget human-object interactions to a humanoid robot and train a reinforcement learning policy to perform these behaviors in simulation. Furthermore, we conduct Sim2Sim transfer to demonstrate the potential of deploying the trained policy to the real humanoid robot. To summarize, we aim to develop methods that enable humanoid robots to learn loco-manipulation skills from diverse human interaction data. We focus on a simplified but representative scenario: lifting and moving a box to a target location.

Method We propose a pipeline that leverages synthesized human-object interaction demonstrations to teach loco-manipulation skills to the Unitree G1 humanoid robot. Our system consists of four key stages. First, we use HiHi Wu et al. (2024), a diffusion-based generative model, to synthesize diverse box-lifting motions given object geometry. Second, synthesized motions are retargeted to the Unitree G1 robot. This maps human joint trajectories into the robot's kinematic space. Finally, we train a reinforcement learning policy in IsaacGym to track both humanoid motion and the object trajectory.

Implementation We used the code from HiHi Wu et al. (2024) to generate box-lifting motions of a human character at the first stage. In the second stage, we first use SMPLX Pavlakos et al. (2019) to fit body shape parameters and scale parameter to match G1 joint position in T-pose. Then we employ the optimized new body shape and original joint rotation data to compute 3D joint positions and use Mink library to conduct inverse kinematics and obtain robot motion. For the motion tracking policy training, we used the codebase from TWIST Ze et al. (2025) and modify observation, rewards, data loading to support tracking global human-object interaction motions.

Results We showcase results of tracking human motion in IsaacGym and Sim2Sim transfer in Mujoco. However, tracking human-object interaction motions is challenging, our current trained policy cannot lift the box sucessfully which need further investigation.

Discussion We found that to achieve global human pose tracking, modifying the reward function is not sufficient. Incorporating additional observations such as next step's reference pose root translation is critical to achieve accurate tracking results. Another interesting finding is that solely relying on rewards designed in graphics field presents obvious instability in feet movement, which can be alleviated through some feet-related rewards designed in the robotics field. Learning to lift a box is not easy, our current trained policy tends to touch the box and then walk forward without successfully moving the box using hands.

Conclusion Our experiments showcase a promising direction to teach humanoid robot motor skills via motion synthesis models. The combination of physics-based character animation in computer graphics field and robotics field provides a potentially scalable approach towards generalized humanoid robots.

Learning Humanoid Loco-Manipulation from Human-Object Interaction

Jiaman Li
Department of Computer Science
Stanford University
jiamanli@stanford.edu

Zhengfei Kuang
Department of Computer Science
Stanford University
zhengfei@stanford.edu

Abstract

We address the problem of learning humanoid loco-manipulation from human-object interactions. Specifically, given human motion and object motion generated by motion synthesis models, we leverage these motion sequences to teach the humanoid robot to imitate the motion that interacts with the given object in simulation. To make our trained policy readily deployable to real robots, we also conduct Sim2Sim transfer to validate the robustness of the learned policy. We conducted extensive experiments to identify effective observation representation and reward functions for successfully imitating the given human-object interactions, showcasing potential of using human-object interaction motions for scalable humanoid loco-manipulation skills learning.

1 Introduction

Loco-manipulation—the combined ability to move through an environment while interacting with objects—is a critical skill for humanoid robots assisting humans in household and manufacturing tasks. Recent advances have shown promising results in learning whole-body controllers that imitate human motion, enabling robots to perform diverse behaviors such as walking, dancing, and kicking. However, these efforts largely neglect object interaction, which is essential for real-world applications. Meanwhile, teleoperation systems have been developed to teach robots object manipulation. These systems use headsets to provide egocentric visual feedback as human operators guide robot arms, often in tabletop settings. While effective for collecting paired egocentric observations and robot actions, these systems are limited: they typically do not capture lower-body movements, and thus cannot support tasks requiring full-body coordination—such as walking while carrying objects or crouching to pick items from the floor. Many real-world tasks—like moving boxes, lifting chairs, or rearranging furniture—require this kind of full-body loco-manipulation. How can we efficiently collect training data for humanoid loco-manipulation? Extending teleoperation to capture full-body motion is tedious, hardware-intensive, and difficult to scale. In contrast, there is a wealth of human-object interaction data—including motion capture, third-person videos, and egocentric recordings—as well as generative models from character animation (e.g., motion diffusion models from language) that can synthesize diverse and scalable training examples. If properly adapted, these can be used to train humanoid loco-manipulation policies. We aim to develop methods that enable humanoid robots to learn loco-manipulation skills from diverse human interaction data. We focus on a simplified but representative scenario: lifting and moving a box to a target location.

Specifically, we propose an approach to teach humanoid how to lift and move a box without collecting data using teleoperation systems. We leverage recent interaction motion synthesis model from the computer graphics research field to generate realistic human motion and object motion in a controllable manner. Then we conduct kinematics-based motion retargeting to obtain robot motions. Based on the reference robot-object interaction motions, we train a reinforcement learning policy to imitate the robot-object interaction motion. We apply domain randomization to improve the robustness of the trained policy. Finally, we conduct Sim2Sim transfer in Mujoco using the trained policy from IsaacGym to validate the robustness of the trained policy.

We showcase successful Sim2Sim transfer using our proposed pipeline. Specifically, given the generated human motion from the interaction synthesis models, we are able to train a robust tracking policy that imitates the global human motion and successfully transfer to Mujoco, demonstrating the potential for Sim2Real transfer. Our experiments show that designing observation representation is critical for training a tracking policy that can accurately track reference motion. Besides, different types of rewards directly affects how the robot behaves in the simulator.

2 Related Work

Humanoid Locomotion In the past two years, humanoid locomotion tasks have been studied extensively. Radosavovic et al. (2024b) formulated the locomotion problem as a next-token prediction problem. Radosavovic et al. (2024a) further explored locomotion in challenging terrain settings. Recently, whole-body motion control methods for humanoid robots have demonstrated robust policy learning for imitating retargeted human motions, enabling dynamic behaviors such as walking and jumping Ji et al. (2024). However, they generally exclude object interactions, limiting their utility in loco-manipulation tasks.

Humanoid Manipulation Teleoperation systems like OpenTeleOperation Cheng et al. (2024) allow human operators to control humanoid arms via egocentric visual feedback from head-mounted displays. These methods have been used to train manipulation policies in tabletop settings. However, they are typically limited to upper-body tracking, since devices like the Apple Vision Pro only provide head and hand pose. Capturing full-body motion would require motion capture suits, making the setup far less scalable. A recent work leverages motion capture to build teleoperation system Ze et al. (2025), enables more tasks involving humanoid locomotion. However, the system requires the operator to wear a mocap suit and actively correct robot behaviors via observing the robot interact. This constraints the efficiency and scalability. In this project, we aim to seek an alternative approach for learning loco-manipulation without human actively involved, which is more scalable and efficient.

Human-Object Interaction in Character Animation Human motion synthesis has been extensively studied in computer graphics research field. With the advent of diffusion models Ho et al. (2020), various motion diffusion models were developed to generate realistic human motion Tevet et al. (2023), human-scene interaction Jiang et al. (2024), and human-object interactions Li et al. (2023b,a); Wu et al. (2024). Recently, HiHi Wu et al. (2024) developed a complete pipeline for long-term human-object interactions and generates realistic motion sequences from language descriptions using diffusion models. These models offer a scalable alternative for producing diverse interaction data without manual demonstrations. Our work bridges motion synthesis and robot learning by transferring synthesized human-object interactions into policies for physical humanoid robots—without relying on teleoperation.

3 Method

We propose a pipeline that leverages synthesized human-object interaction demonstrations to teach loco-manipulation skills to the Unitree G1 humanoid robot as shown in Figure 1. Our system consists of four key stages. First, we use HiHi Wu et al. (2024), a diffusion-based generative model, to synthesize diverse box-lifting motions given object geometry. Second, synthesized motions are retargeted to the Unitree G1 robot. This maps human joint trajectories into the robot's kinematic space. Finally, we train a reinforcement learning policy in IsaacGym to track both humanoid motion and the object trajectory and conduct Sim2Sim transfer in Mujoco. We elaborate each step as follows.

Motion Synthesis. HiHi Wu et al. (2024) developed a multi-stage approach to generate human-object interaction motions including full-body human motion, finger motion and object motion. Specifically, given start human pose, start object pose, and a language descripotion such as "Lift the box, move the box and put it down.", the first stage generates coarse full-body human motion and object motion. Then we can divide the whole sequence into pre-contact, contact, and post-contact

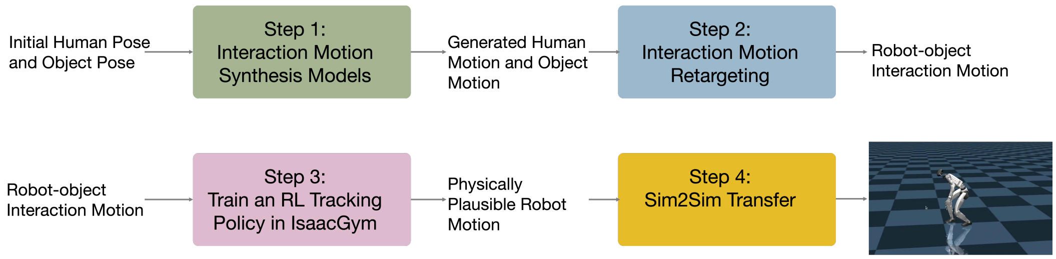

Figure 1: Method Overview. Our approach consists of four steps. Starting from a generated human-object interaction sequence, we conduct motion retargeting to obtain kinematics-based robot motions. Then we train an RL policy to imitate human-object interactions. Finally, we test Sim2Sim transfer, validting the robustness of the trained policy.

phases based on wrist-object distance. We compute an average wrist pose during contact phase as iniitialization for Stage 2 Grasp Pose Optimization. In Stage 2, HiHi Wu et al. (2024) employ DexGraspNet Wang et al. (2023) to optimize finger pose given an initial wrist pose. The optimization incorporated contact terms and penetration terms to ensure the optimized grasp pose is realistic. Stage 3 uses the optimized grasp pose as addtional condition for the motion diffusion model to re-generate human motion and object motion that satisfy the grasp pose constraints. Stage 4 generates finger motion for pre-contact and post-contact phases. This motion synthesis step provides realistic human motion and object motion.

Motion Retargeting. The first step provides human-object interaction motion. However, the motion is in SMPLX format and cannot be directly used for Unitree G1 robots. In our second step, we conduct motion retargeting to obtain robot motions. We first use SMPLX Pavlakos et al. (2019) to fit body shape parameters and scale parameter to match G1 joint position in T-pose. Then we employ the optimized new body shape and original joint rotation data to compute 3D joint positions and use Mink library to conduct inverse kinematics and obtain robot motion.

RL Tracking Policy Training. We use IsaacGym as the physics engine to perform simulation. The simulated character is skeleton-driven where each joint is modeled as a motor controlled through a PD controller. The control policy works at 50Hz, while the simulation runs at 500Hz. The G1 character has 38 body links and 29 controllable joints. The state space $\mathbf{s}_t \in \mathbb{R}^d$ consists of root orientation, root linear velocity, root angular velocity, dof, dof velocity, the position, orientation and linear and angular velocities of the character's hand links and the manipulating objects $\mathcal{O}$. To track the target motion, additional observation $\mathbf{o}_t \in \mathbb{R}^{d'}$ including the target position and orientation of object and target root translation is also introduced as the input to the control policy. The action space is $\mathbf{a}_t \in \mathbb{R}^{29}$. Our control policies are optimized using PPO as the backbone reinforcement learning algorithm plus some techniques introduced in RMA Kumar et al. (2021) following Ze et al. (2025).

Besides the reward introduced in Ze et al. (2025), to adapt the codebase to track global human motion and object motion, we define additional reward functions containing three terms to evaluate the full-body pose and object pose separately:

$$r = 0.45 r_{\text{body}} + 0.4 r_{\text{object}} + 0.15 r_{\text{relative}}, \tag{1}$$

where r_{body} evaluates the full-body poses, r_{relative} evaluates the hand poses related to the target manipulating object, and r_{object} evaluates the object trajectory.

Sim2Sim Transfer. To evaluate the robustness of the trained policy and prepare for Sim2Real transfer, Sim2Sim transfer is a critical step. We trained an RL tracking policy on IsaacGym and tested the trained policy on Mujoco. This Sim2Sim step involves properly resetting initial character state on Mujoco and conducting some data representation conversion between Mujoco and IsaacGym. Essentially, we need to use the same observation representation as in IsaacGym to predict action, and apply action using Mujoco's dynamics. These two simulators require different simulation frequency.

4 Experimental Setup

We used the code from HiHi Wu et al. (2024) to generate box-lifting motions of a human character at the first stage. In the second stage, we first use SMPLX Pavlakos et al. (2019) to fit body shape parameters and scale parameter to match G1 joint position in T-pose. Then we employ the optimized new body shape and original joint rotation data to compute 3D joint positions and use Mink library to conduct inverse kinematics and obtain robot motion. For the motion tracking policy training, we used the codebase from TWIST Ze et al. (2025) and modify observation, rewards, data loading to support tracking global human-object interaction motions.

We conducted experiment in two different settings. The first setting is tracking human motion only. This setting is used to establish basic understanding about the effects of different rewards, different observation representations. And this step is critical for tracking human-object interaction motion. The second setting is tracking both human motion and object motion. We need to add object motion related rewards and observation based on the previous setting. Our evaluation mainly rely on visually checking the tracked human motion and corresponding reference motion.

5 Results

We showcase results of tracking human motion in IsaacGym and Sim2Sim transfer in Mujoco in Figure 2 and Figure 3. Tracking human-object interaction motions is challenging, our current trained policy struggle to lift the box successfully while maintaining realistic human motion, which need further investigation. We showcased some results of failed lifting boxes in Figure 4. We observed two scenarios. One is the robot can follow the reference motion well, but ignoring the object. The other one is the robot stay at the initial position and only track local human pose well. We further investigated the issue and found a problem introduced by the reference object pose. The reason of failing to lift the box might be the contact position is not realistic in the reference motion. To lift the box, the hands of a robot should be in contact with the opposite sides of the box instead of adjacent sides. After fixing this issue, we removed all the robotics rewards and only kept the object motion and human motion matching reward, and we can obtain a policy that lifts a box successfully. However, the human motion has some feet flapping issues 5.

5.1 Qualitative Analysis

We first trained a tracking policy to imitate human motion only. Based on TWIST Ze et al. (2025), we added next step's root translation as part of the observation input, and we added reward for global joint position tracking and global joint orientation tracking so that the character can follow the root trajectories in the reference motion. We found that root translation as observation is important for accurate global root trajectory tracking.

To conduct Sim2Sim transfer, we also adapted previous codebase to our setting by using the first frame as initialization. TWIST Ze et al. (2025) does not rely on first frame initialization since it only take the reference local pose as input, while resetting the first frame using reference's first pose is important in our setting.

We further include object motion into our RL policy training. This requires adding additional object state as part of observation, and adding reference object pose as input to predict robot actions. Besides, we added object motion matching reward to encourage the output actions produce proper torques so that the box can be lifted and follow the reference object trajectory. We have tried many different settings to balance different reward terms, however, we still cannot get a policy to sucessfully lift the box up at this moment. We will investiagte the problem in the future work.

6 Discussion

We found that to achieve global human pose tracking, modifying the reward function is not sufficient. Incorporating additional observations such as next step's reference pose root translation is critical to achieve accurate tracking results. Another interesting finding is that solely relying on rewards designed in graphics field presents obvious instability in feet movement, which can be alleviated through some feet-related rewards designed in the robotics field. Learning to lift a box is not easy,

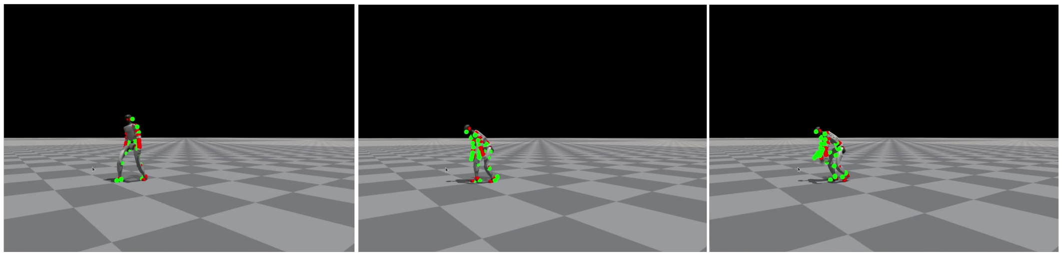

Figure 2: Visualization of results.

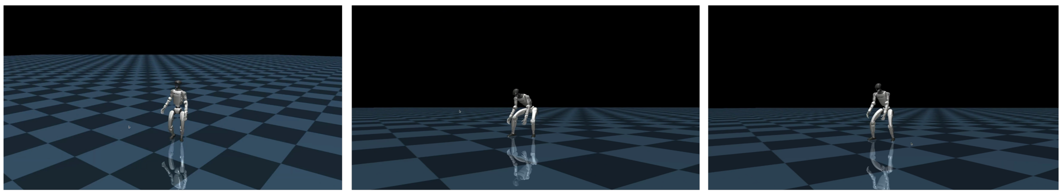

Figure 3: Visualization of Sim2Sim transfer results.

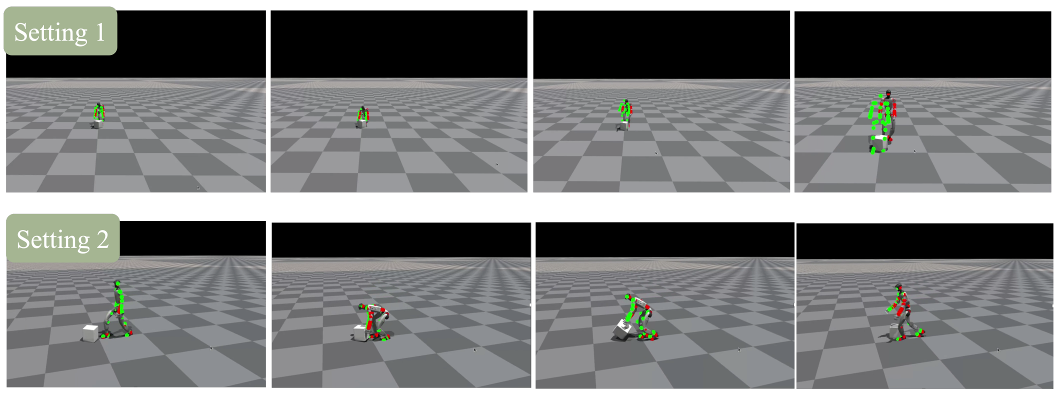

Figure 4: Visualization of HOI tracking results. Current trained policy can track human motion only, but cannot successfully lift the box up. This is due to reference object motion issue. The contact position is not realistic.

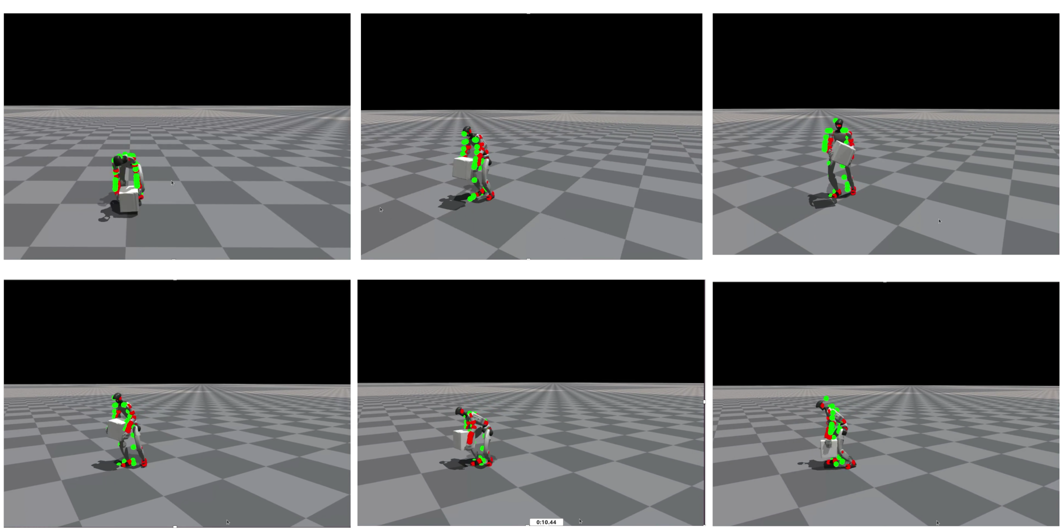

Figure 5: Visualization of HOI tracking results. This trained policy can lift the box up.However, there exists feet flapping artifacts in the tracked human motion.

our current trained policy tends to touch the box and then walk forward without successfully moving the box using hands. After fixing the reference box motion issue, we are able to get a policy that can lift the box and move the box, however, the human motion does not realistic and contain obvious artifacts. If we add both robotics reward and motion matching reward, we found there would be a wrist pose mismatch and lead to failure lifting behaviors. We think this might be that the reference motion does not satisfy some dof limits in G1. We will investigate the issue of the refeerence motion in the future.

7 Conclusion

In this project, we studied the problem of teaching humanoid robot loco-manipulation skills from human-object interaction demonstrations. Specifically, we employ a motion synthesis model in computer graphics domain to generate human motion and object motion, specifically a human lifts a box, and we use the generated interaction motion to conduct motion retargeting and train an RL policy to imitate the interaction motion for humanoid robots in simulation. Our experiments showcase a promising direction to teach humanoid robot motor skills via motion synthesis models. The combination of physics-based character animation in computer graphics field and robotics field provides a potentially scalable approach towards generalized humanoid robots.

8 Team Contributions

- **Group Member 1 (Jiaman Li):** 50% work including Step 1 (Motion Synthesis) and Step 3 (RL policy training).
- **Group Member 2 (Zhengfei Kuang):** 50% work including Step 2 (Motion Retargeting) and Step 4 (Sim2Sim Transfer).

Changes from Proposal We followed the proposed approach in our proposal. However, due to time constraints, we only achieved sucessful human motion tracking. There still exist some issues with tracking human-object interaction. We have some promising resuslts for lifting the box, however, feet flapping issues were not solved.

References

Xuxin Cheng, Jialong Li, Shiqi Yang, Ge Yang, and Xiaolong Wang. 2024. Open-television: Teleoperation with immersive active visual feedback. *CoRL* (2024).

Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems (NeurIPS)* 33 (2020), 6840–6851.

Mazeyu Ji, Xuanbin Peng, Fangchen Liu, Jialong Li, Ge Yang, Xuxin Cheng, and Xiaolong Wang. 2024. Exbody2: Advanced expressive humanoid whole-body control. *arXiv preprint arXiv:2412.13196* (2024).

Nan Jiang, Zimo He, Zi Wang, Hongjie Li, Yixin Chen, Siyuan Huang, and Yixin Zhu. 2024. Autonomous Character-Scene Interaction Synthesis from Text Instruction. *arXiv preprint arXiv:2410.03187* (2024).

Ashish Kumar, Zipeng Fu, Deepak Pathak, and Jitendra Malik. 2021. Rma: Rapid motor adaptation for legged robots. *arXiv preprint arXiv:2107.04034* (2021).

Jiaman Li, Alexander Clegg, Roozbeh Mottaghi, Jiajun Wu, Xavier Puig, and C Karen Liu. 2023a. Controllable human-object interaction synthesis. *arXiv preprint arXiv:2312.03913* (2023).

Jiaman Li, Jiajun Wu, and C Karen Liu. 2023b. Object Motion Guided Human Motion Synthesis. *ACM Trans. Graph.* 42, 6 (2023).

Georgios Pavlakos, Vasileios Choutas, Nima Ghorbani, Timo Bolkart, Ahmed AA Osman, Dimitrios Tzionas, and Michael J Black. 2019. Expressive body capture: 3d hands, face, and body from a single image. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10975–10985.

Ilija Radosavovic, Sarthak Kamat, Trevor Darrell, and Jitendra Malik. 2024a. Learning humanoid locomotion over challenging terrain. *arXiv preprint arXiv:2410.03654* (2024).

Ilija Radosavovic, Bike Zhang, Baifeng Shi, Jathushan Rajasegaran, Sarthak Kamat, Trevor Darrell, Koushil Sreenath, and Jitendra Malik. 2024b. Humanoid locomotion as next token prediction. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*.

Guy Tevet, Sigal Raab, Brian Gordon, Yonatan Shafir, Amit H Bermano, and Daniel Cohen-Or. 2023. Human Motion Diffusion Model. In *International Conference on Learning Representations (ICLR)*.

Ruicheng Wang, Jialiang Zhang, Jiayi Chen, Yinzhen Xu, Puhao Li, Tengyu Liu, and He Wang. 2023. Dexgraspnet: A large-scale robotic dexterous grasp dataset for general objects based on simulation. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 11359–11366.

Zhen Wu, Jiaman Li, Pei Xu, and C Karen Liu. 2024. Human-object interaction from human-level instructions. *arXiv preprint arXiv:2406.17840* (2024).

Yanjie Ze, Zixuan Chen, JoÃĢo Pedro AraÃšjo, Zi-ang Cao, Xue Bin Peng, Jiajun Wu, and C Karen Liu. 2025. TWIST: Teleoperated Whole-Body Imitation System. *arXiv preprint arXiv:2505.02833* (2025).